

SINGLE RESPONSIBILITY PRINCIPLE

MISSUNDERSTOOD AND UNDERESTIMATED

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INTRO


Active poll
How familiar are you with the topic - Single Responsibility Principle?
I'm an expert
0%
I have some solid background
0%
I have some basic knowledge
0%
I'm completely green
0%
Join at
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SINGLE RESPONSIBILITY PRINCIPLE

"A class should have only one reason to change."

Robert C. Martin,"Agile Software Development: Principles, Patterns, and Practices", 2002

SINGLE RESPONSIBILITY PRINCIPLE

IS ONE OF THE SIMPLEST OF THE PRINCIPLES, AND ONE OF THE HARDEST TO GET



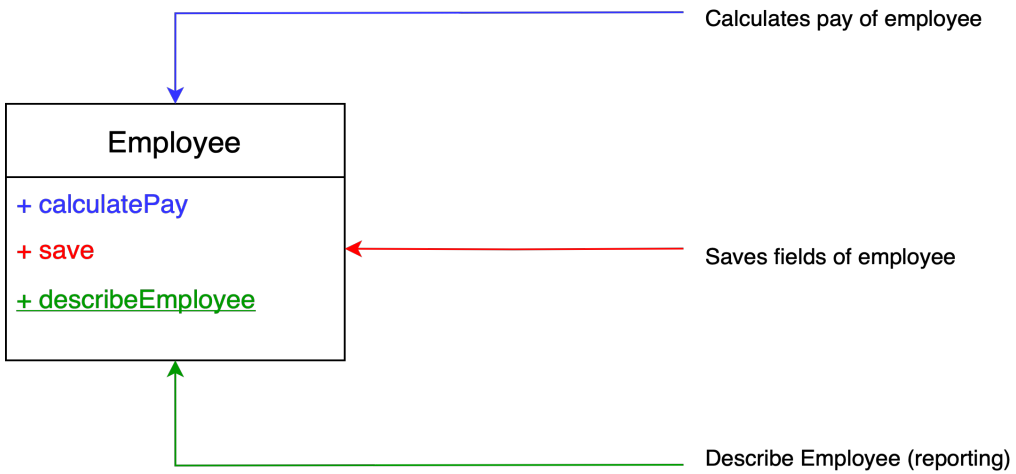
SINGLE RESPONSIBILITY PRINCIPLE (SRP)

"In context of SRP, we define
a responsibility
to be
a reason for change."

Robert C. Martin,"Agile Software Development: Principles, Patterns, and Practices", 2002

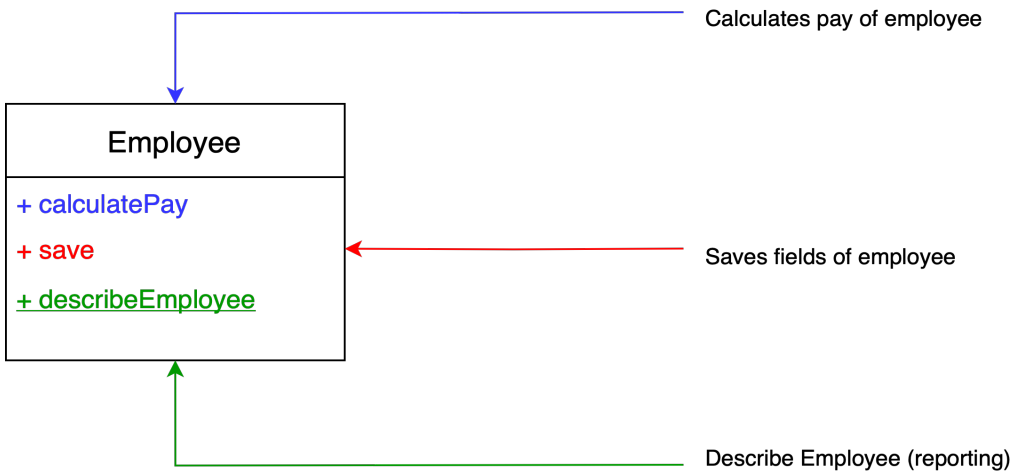
SINGLE RESPONSIBILITY PRINCIPLE

"A CLASS SHOULD HAVE ONLY ONE REASON TO CHANGE." - ROBERT C. MARTIN



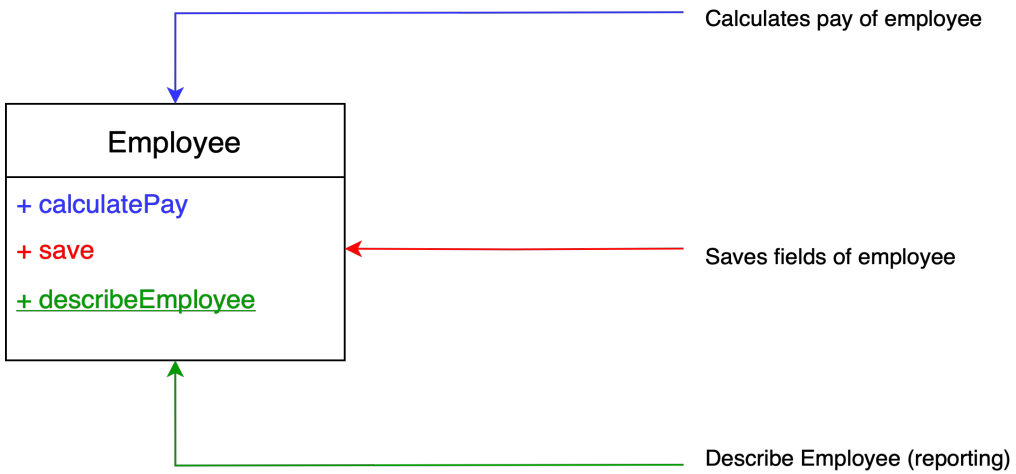
RESPONSIBILITY

How many responsibilities does this class have?

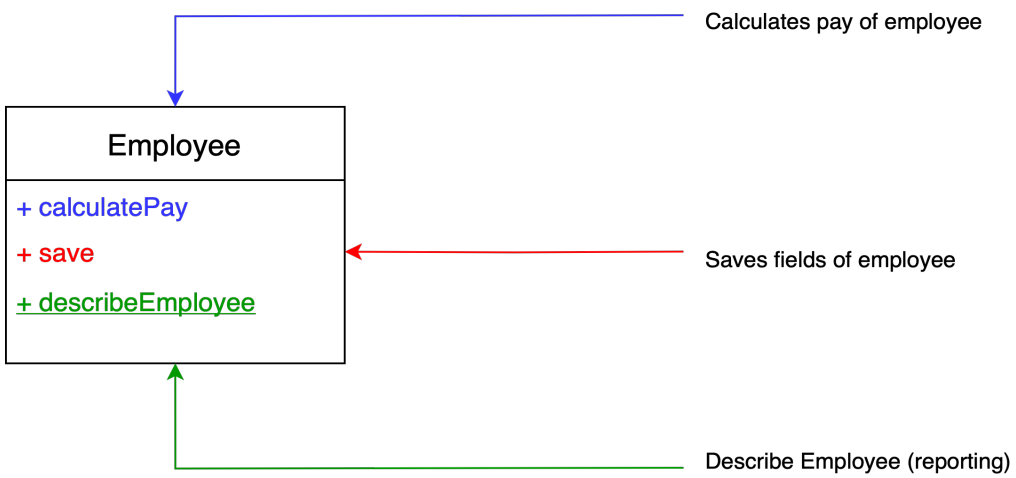


RESPONSIBILITY

How many responsibilities does this class have? 3



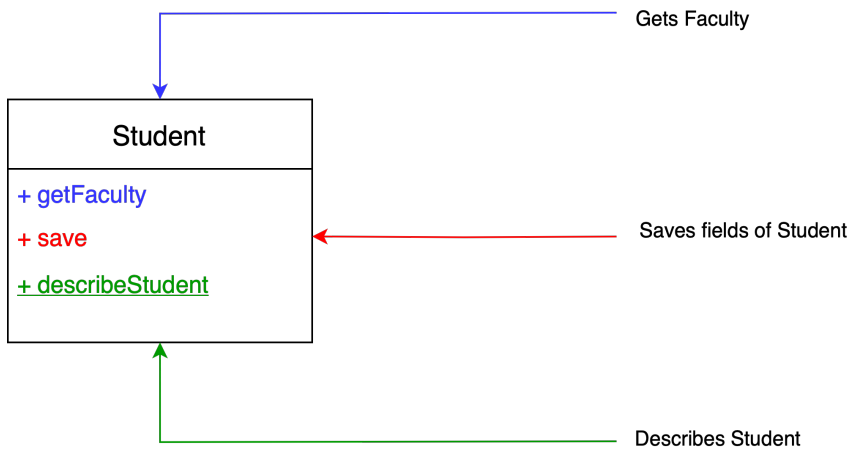
CONCLUSION – VIOLATES SRP



- The class knows about
  - business rules
  - databases
  - reports and formatting
- Responsibilities > 1
- Each of these responsibilities will cause the employee class to use other classes in the system.

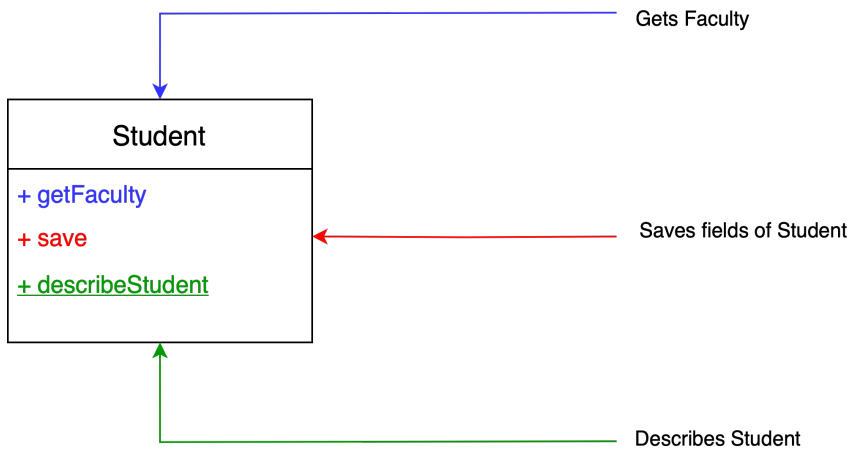
RESPONSIBILITY

How many responsibilities does this class have?



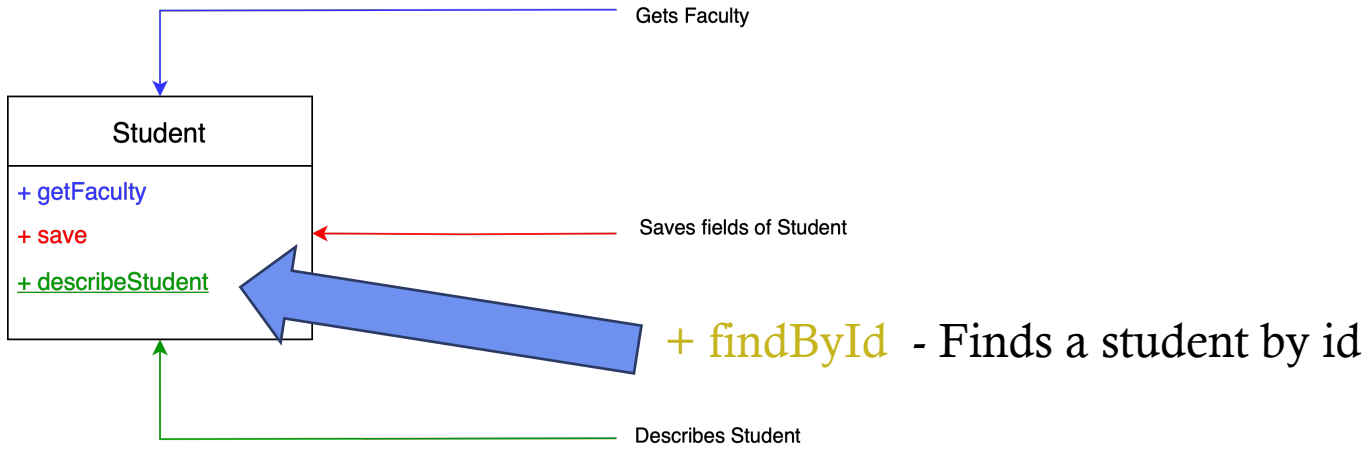
RESPONSIBILITY

How many responsibilities does this class have? 3



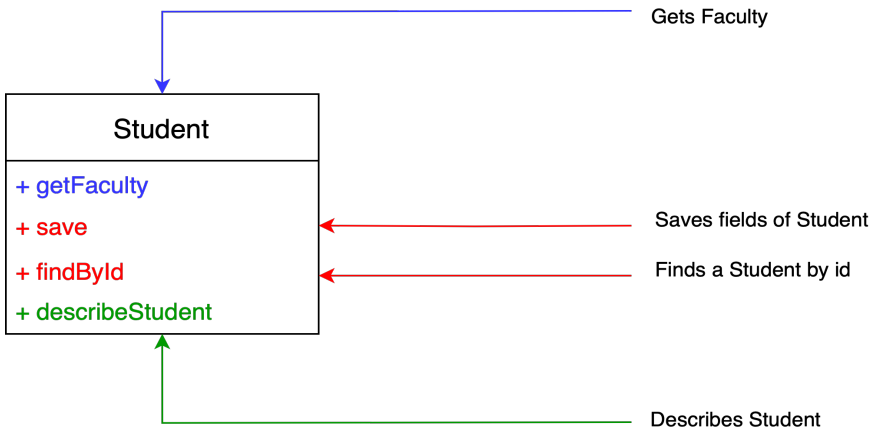
RESPONSIBILITY

Let us add a new function to the class.
How many responsibilities does this class have?



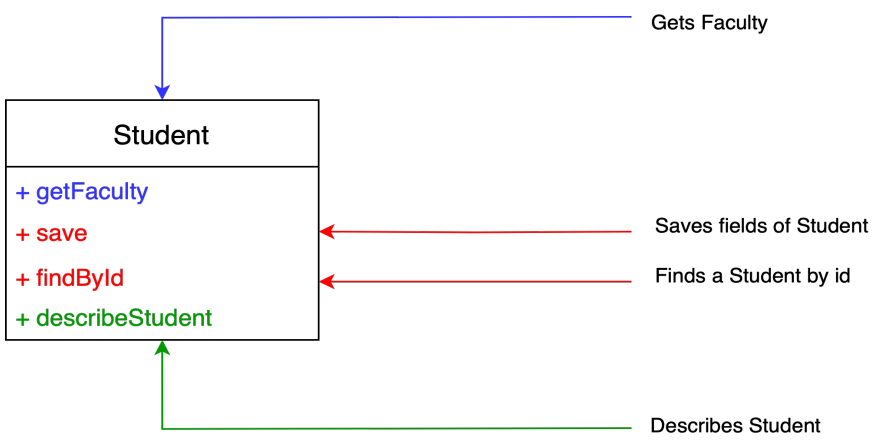
RESPONSIBILITY

How many responsibilities does this class have? 3



same family as save, both database functions

CONCLUSION – VIOLATES SRP





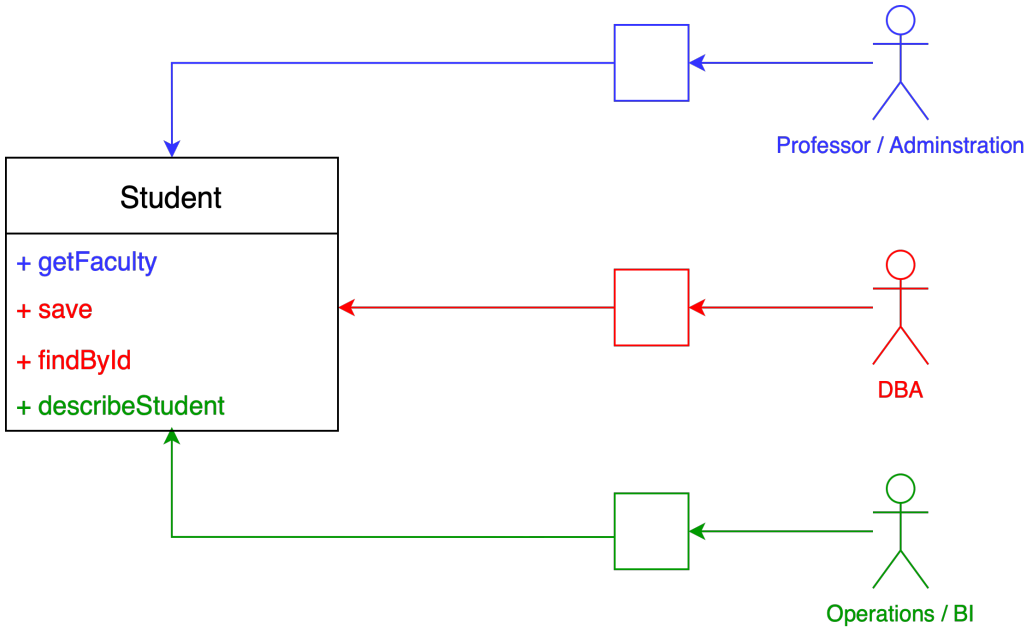
- The class knows about
  - business rules
  - databases
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- Each of these responsibilities will cause the student class to use other classes in the system.

RESPONSIBILITY – SOURCE OF CHANGE

- The SRP is about users
- Who are the users, who add changes to that method?

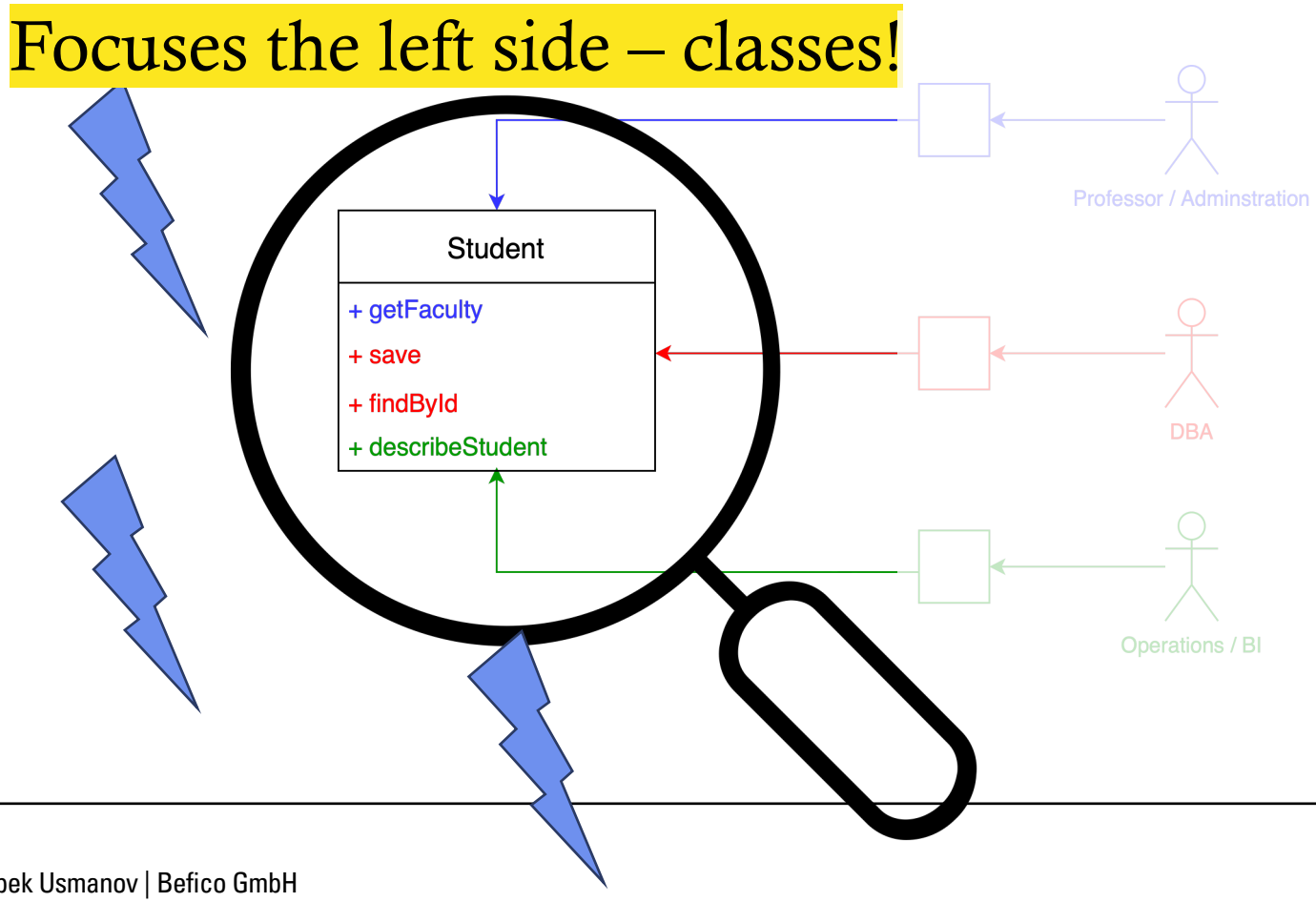
RESPONSIBILITY

WHO ARE THE PEOPLE, WHO ARE THE SOURCES OF CHANGE FOR THE STUDENT CLASS?



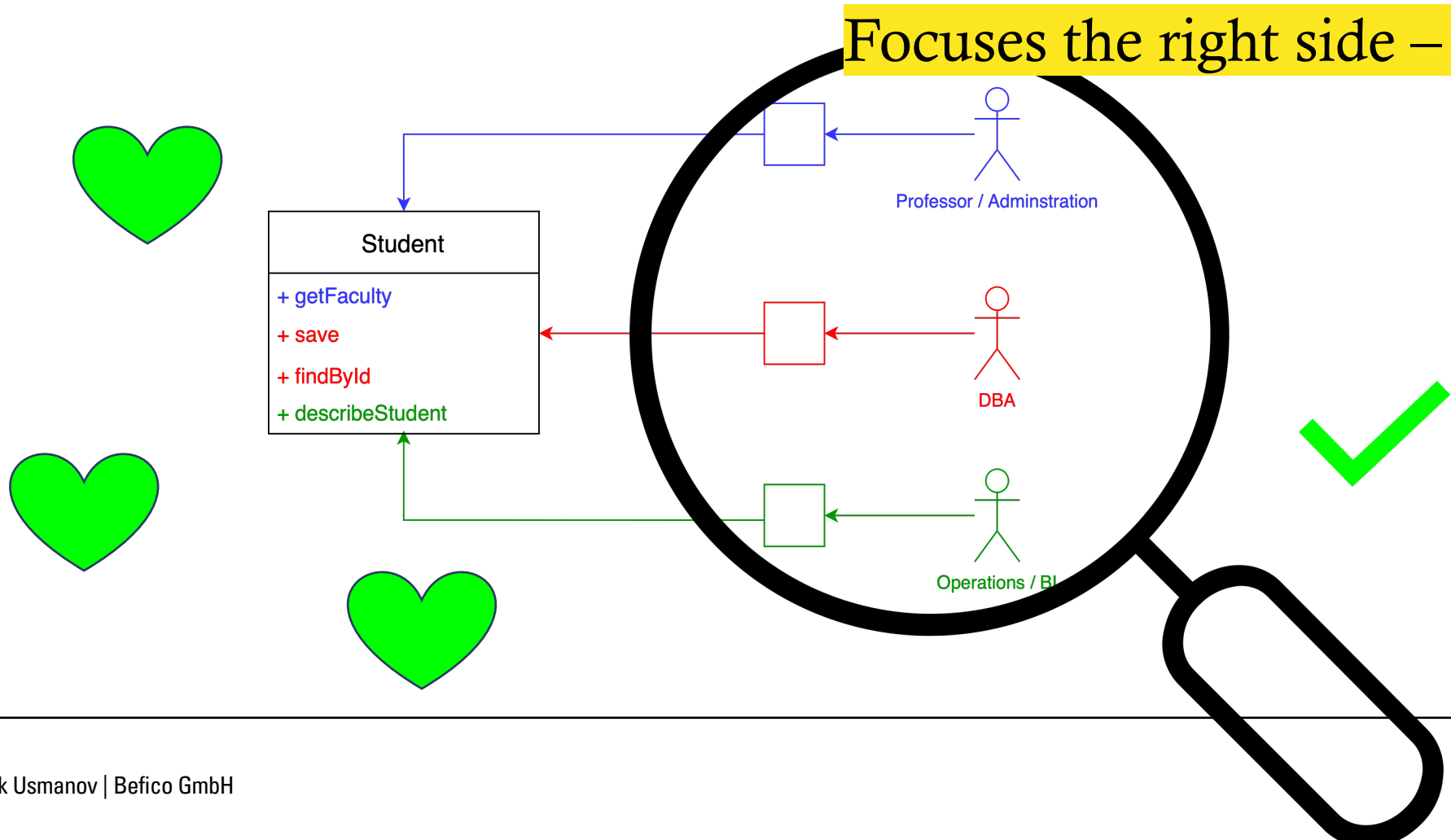
SINGLE RESPONSIBILITY PRINCIPLE

"A CLASS SHOULD HAVE ONLY ONE REASON TO CHANGE." - ROBERT C. MARTIN



SINGLE RESPONSIBILITY PRINCIPLE

RESPONSIBILITY IS A FAMILY OF FUNCTIONS THAT SERVES ONE PARTICULAR ACTOR



REPHRASE

SINGLE RESPONSIBILITY PRINCIPLE

"A class should have only one reason to change."

Robert C. Martin, 2002

SINGLE OWNERSHIP PRINCIPLE

"A class should serve only one owner, who defines the requirements to create or change it."

Nodirbek Usmanov, 2023

REPHRASE 2

SINGLE RESPONSIBILITY PRINCIPLE

"A class should have only one reason to change."

Robert C. Martin, 2002

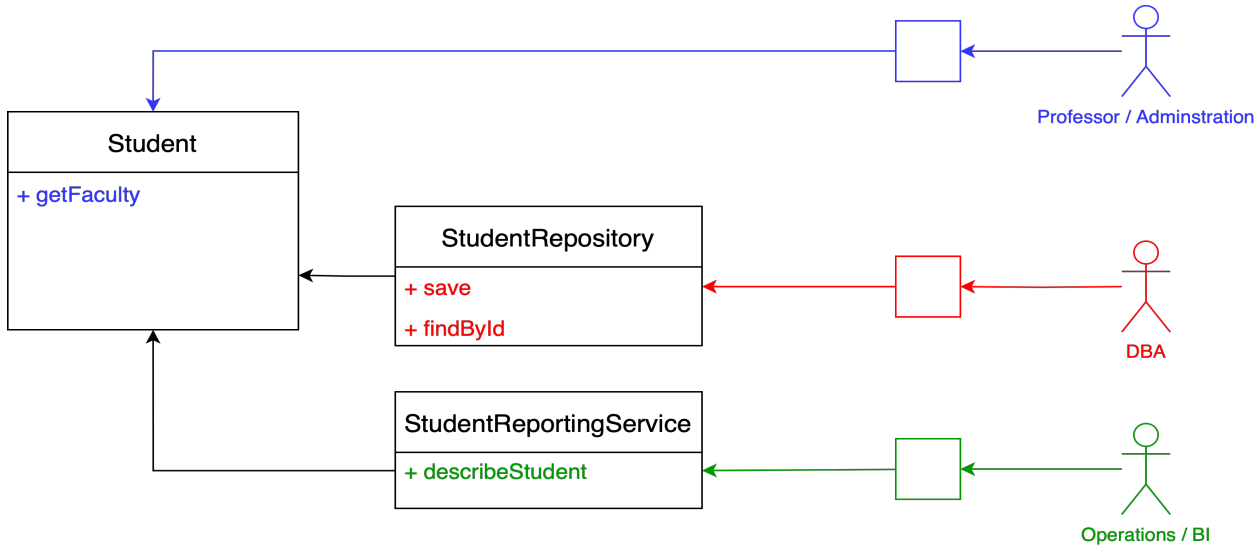
SINGLE OWNERSHIP PRINCIPLE

"A class should serve only one owner, who defines a family of functions (responsibility)."

Nodirbek Usmanov, 2023

REFACTOR

AWARE OF TRANSITIVE DEPENDENCIES



TAKE AWAY

- A responsibility is a family of functions that serves one particular actor
- Increase the cohesion between things that change for same reasons
- Decrease the coupling between those things that change change for different reasons

LINKS

- The Single Responsibility Principle (08 May 2014), by Robert C. Martin (Uncle Bob), https://blog.cleancoder.com/uncle-bob/2014/05/08/SingleReponsibilityPrinciple.html
- Agile Software Development: Principles, Patterns, and Practices, 2002, by Robert C. Martin (Uncle Bob)

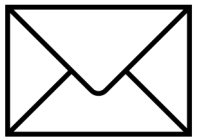
THANK YOU



Reviewer Ulugbek Usmanov
CGI Consulting Deutschland



Reviewer Marcus Schroeder
flint & sparks GmbH, Berlin



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